

particular service associated with the network.

Geolocation extraction

The real-time location of an IP address can be extracted using Python and Google APIs with the use of the *pygeoip* module. First of all, import the *GeoIP* database from the URL <http://dev.maxmind.com/geoip/legacy/geolite/>.

Once the database is loaded and mapped with the Python installation, any IP address can be scanned with global visibility and location.

```
>>> import pygeoip
>>> myGeoIP = pygeoip.GeoIP('GeoIPDataSet.dat')
>>> myGeoIP.country_name_by_addr('<IP Address>')
'United States'
```

To look up the country, use the following commands:

```
>>> myGeoIP = pygeoip.GeoIP('GeoIPDataSet.dat')
>>> myGeoIP.country_code_by_name('google.com')
'US'
>>> myGeoIP.country_code_by_addr('<IP Address>')
'US'
>>> myGeoIP.country_name_by_addr('<IP Address>')
'United States'
```

To look up the city, use the following commands:

```
>>> myGeoIP = pygeoip.GeoIP('GeoIPCity.dat')
>>> myGeoIP.record_by_addr('<IP Address>')
{
    'city': u'Mountain View',
    'region_code': u'CA',
    'area_code': 550,
    'time_zone': 'America/Los_Angeles',
    'dma_code': 807,
    'metro_code': 'San Francisco, CA',
    'country_code3': 'USA',
    'latitude': 38.888222,
    'postal_code': '94043',
    'longitude': -123.37383,
    'country_code': 'US',
    'country_name': 'United States',
    'continent': 'NA'
}
>>> myGeoIP.time_zone_by_addr('<IP Address>')
'America/Los_Angeles'
```

Real-time extraction from social media

The live and real-time data from social media platforms can be downloaded using Python scripts. In Python, there are many modules and extensions with which the interfacing with WhatsApp, Twitter, Facebook, LinkedIn and many



Figure 6: Downloadable databases for GeoIP mapping

Downloads					
Database	Download links				
	Binary / gzip	Binary / xz	CSV / gzip	CSV / zip	CSV / xz
GeoLite Country	Download	Gzip only	Zip only	Download	Zip only
GeoLite Country IPv6	Download	Gzip only	Download	Gzip only	Zip only
GeoLite City	Download	Download	Zip and xz only	Download	Download
GeoLite City IPv6 (Beta)	Download	Gzip only	Download	Gzip only	Gzip only
GeoLite ASN	Download	Gzip only	Zip only	Download	Zip only
GeoLite ASN IPv6	Download	Gzip only	Zip only	Download	Zip only

The GeoLite Legacy databases may also be downloaded and updated with our [GeoIP Update program](#).

Figure 7: Download links for GeoIP databases with IPv6 compatibility

other platforms can be done.

Python package index (PyPi)

PyPi (<https://pypi.python.org>) is the software repository of enormous Python packages for interfacing with other platforms. PyPi is freely available for Python developers without any licensing or subscriptions.

You can download the Python codes for the three actions given below from http://opensourceforu.com/article_source_code/sept16/digital_forensic.zip

1. Fetching the list of followers from Twitter about any user
2. Fetching the Twitter timeline for any user name
3. Real-time extraction of live tweets from Twitter

With the execution of the Python code downloaded from the above link, the live real-time discussion on any topic or keyword can be fetched. With this execution, the real-time discussion on the word 'India' will be fetched along with the details on the users involved in the transmission and distribution of tweets. The user data will include the user name, device, tweet, followers' list, timestamp of the tweet, platform used, etc. **END** 

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